非线性模型

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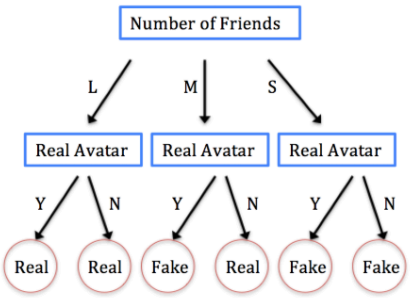
Tags	

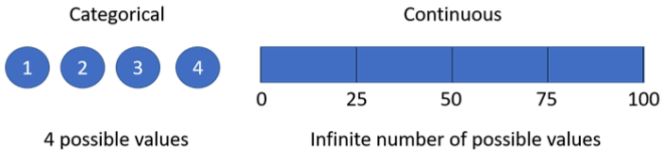
Class 23 Nonlinear Models

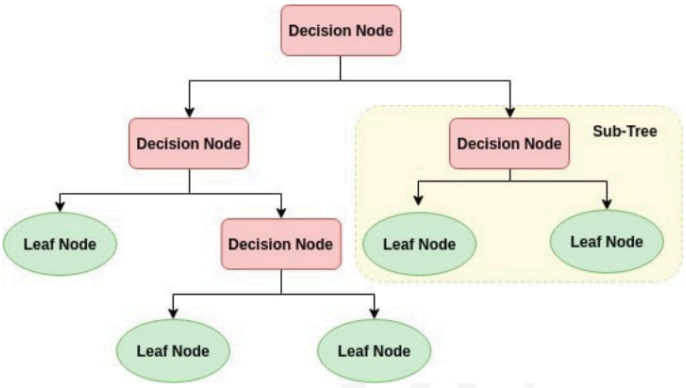
1. Decision Tree



1.1. What is Decision Tree

- Decision Tree is a supervised prediction model.
- Decision Tree is usually a k-nary tree. Each leaf denotes one classification or continuous value of the dependent variable.
- Two types of Decision Tree:
  - Classification Tree
  - Regression Tree

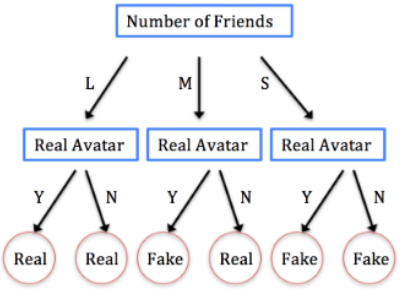




Before we dive deep, let's get familiar with some of the terminologies:

1. **Root Node** (Top Decision Node): It represents the entire population or sample and this further gets divided into two or more homogeneous sets.
2. **Splitting**: It is a process of dividing a node into two or more sub-nodes.
3. **Decision Node**: When a sub-node splits into further sub-nodes, then it is called a decision node.
4. **Leaf/ Terminal Node**: Nodes with no children (no further split) is called Leaf or Terminal node.
5. **Pruning**: When we reduce the size of decision trees by removing nodes (opposite of Splitting), the process is called pruning.
6. **Branch / Sub-Tree**: A sub section of the decision tree is called branch or sub-tree.
7. **Parent and Child Node**: A node, which is divided into sub-nodes is called a parent node of sub-nodes whereas sub-nodes are the child of a parent node.

1.2. How to Build a Decision Tree



- Two steps:
  - **Feature prioritization**
  - Feature Splitting

- (扩展内容) Algorithm Details - ID3 Algorithm
  - At each iteration step, we split the feature which can give us the largest entropy gain.
  - Contribution of each component in the system to the system Entropy(熵):
    Entropy of the entire information system:

$$H = \sum_{i=1}^{n} p_i * I_e(i) = -\sum_{i=1}^{n} p_i log_2 p_i$$

  -
  - Example

Independent Variables			Dependent Variables
Real Avatar (A)	Friends (B)	Blog (C)	Real User Account (U)
N	S	S	N
Y	L	S	Y
Y	M	L	Y
Y	M	M	Y
Y	M	L	Y
N	L	M	Y
N	S	M	N
N	M	L	Y
N	S	M	Y
Y	S	S	N

0.879 (本身的熵)

$H_{Y|C}$ = 0.603 -> gain is 0.276 when C

$H_{Y|A}$ = 0.846 -> gain is 0.033 when A

$H_{Y|B}$ = -> gain is 0.553 when B

Entropy of the current system:
系统本身的熵

$$H = \sum_{i=1}^{n} p_i * I_e(i) = -\sum_{i=1}^{n} p_i log_2 p_i$$

= Yes的熵 + No的熵

$$= -p_Y log_2 p_Y + (-p_N log_2 p_N)$$

P(Y)=0.7 = p,i=y
P(N)=0.3

$$H_U = -\underset{Y}{0.7log_2 0.7} - \underset{N}{0.3log_2 0.3} = 0.879$$

Information gain = Entropy - Conditional Entropy

Updated conditional entropy after adding feature C into system:

Conditional Entropy:
加入feature C以后系统的熵
$H_{Y|C}$ = P(C=L) * H(Y | C=L) + P(C=M) * H(Y | C=M) + P(C=S) * H(Y | C=S)

1) Step 1， 计算P
P(C=L) = 0.3
P(C=M) =0.4
P(C=S)= 0.3

2) Step 2，就算H
因为：

$$H = \sum_{i=1}^{n} p_i * I_e(i) = -\sum_{i=1}^{n} p_i log_2 p_i = \text{Yes的熵 + No的熵}$$

$$= -p_Y log_2 p_Y + (-p_N log_2 p_N)$$

所以，所有的概率变成了条件概率，因为有了C：

H(Y | C=L) = $-p_{(Y|C=L)} log_2 p_{(Y|C=L)} + (-p_{(N|C=L)} log_2 p_{(Y|C=L)})$

P(Y|C=L）= 3/3， P(N|C=L)=0/3=0
P(Y|C=M）= 3/4， P(N|C=M)=¼
P(Y|C=S）= 1/3， P(N|C=S)=2/3

3) Step 3, combine Step 1 and Step 2， 带回总公式：

$$H_{Y|C} = \underset{L}{0.3} * (-\underset{L+N}{\tfrac{0}{3}log_2\tfrac{0}{3}} - \underset{L+Y}{\tfrac{3}{3}log_2\tfrac{3}{3}}) + \underset{M}{0.4} * (-\underset{M+N}{\tfrac{1}{4}log_2\tfrac{1}{4}} - \underset{M+Y}{\tfrac{3}{4}log_2\tfrac{3}{4}}) + \underset{S}{0.3} * (-\underset{S+N}{\tfrac{2}{3}log_2\tfrac{2}{3}} - \underset{S+Y}{\tfrac{1}{3}log_2\tfrac{1}{3}}) = 0.603$$

4）Step 4, 计算新引入feature对系统的造成影响
The information gain(增益，就是变化) by splitting feature C = 0.879-0.603 = 0.276

5）同样的方法，分别计算A引入和B引入造成的影响
The entropy gain by splitting feature A = 0.033
The entropy gain by splitting feature B = 0.553.

引入B的增益最大，Therefore, we choose Feature B as the **first** splitting feature.
之后的其他的分支，就在选定第一个的情况下计算。

More details to read - Other metrics of Decision Tree: ***https://www.geeksforgeeks.org/machine-learning/decision-tree-algorithms/***

1.3. Compare Decision Tree with Linear Regression

- We can formulate decision tree model as the following formula:

$$f(X) = \sum_{k=1}^{K} contribution(x_k)$$

这里面contribution(x_k)表示的是输入数据第k个feature对输出结果产生的作用, 它可能是信息熵的减小, 也可能是Gini Impurity的减小. 把上面的公式与Linear regression公式相比较:

$$y = f(X) = b + \sum_{k=1}^{K} a_k x_k$$

在Linear regression中, x_k这一项对f(X)的contribution是确定的, 只与自己的取值和对应的系数有关. 而Decision Tree中contribution(x_k)的取值, 不仅仅与x_k有关, 也与X中其他feature所决定的整个Decision Path有关(更具体地说, feature在path上的先后顺序不同, 它的contribution就会不同). 很明显, 这里是一种非线性关系.

Decision tree:
优点：他可以用来描述x和y之间更加复杂的关系，建模的能力强
缺点：严重overfitting

（了解一下）解决Decision Tree的overfitting问题的方法：Pruning (剪枝)
1. Post-pruning
2. Pre-pruning

2. Ensemble learning

In ensemble learning, we train multiple learners to solve the same problem.
Ensemble learning including Boosting method and Bagging method.
【Advanced topic】
- **Bagging -> focus on reducing variance**
  - **Learn independently from each other in parallel**
  - **Combine with deterministic averaging process**
- Boosting -> focus on reducing bias
  - Learn sequentially in a adaptative way (a base model depends on the previous ones)
  - Combine with deterministic strategy

3. Random Forest

3.1. Bagging

- Bagging == bootstrap aggregating
- Bagging is an ensemble learning method.
- Bagging method - sampling with replacement.
- Advantage of Bagging method:
  Decrease variance by introducing randomness into your model framework.
- Random Forest leverages a bagging method to solve overfitting problems in complete decision trees.

(拓展）Bagging impact on variances:

$$variance\ after\ bagging = \rho * \sigma^2 + (1 - \rho) * \sigma^2 / n$$

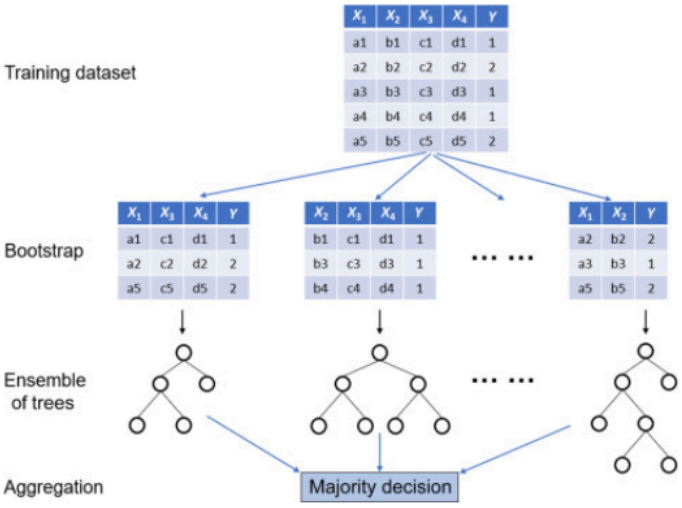
where ρ is the correlation coefficient in between datasets, σ^2 is the original variance and n is the number of small weak models.

More details to read: https://arxiv.org/html/2402.12668v1

3.2. Description of Random Forest

假设总体数据中有N个样本, 每个样本有m个特征属性(feature的个数是m个).
- 每次从原数据中有放回地随机选取N个观测值(注意这个过程中有的样本可能被选择了多次, 有的样本可能被没有被包含在取出的N个观测值中)
- 从得到的这些样本中随机选取k (k<m)个特征, 利用这k个特征计算出一个最佳分裂方式的decision tree.
- 多次重复上面的操作, 得到一组互不相同的decision tree, 把它们整合起来构成random forest模型, 对新数据进行分类或者预测.

The recommend k 一般是 floor(sqrt(m))



3.3. Summary of Random Forest

- Sampling idea in Random Forest algorithm:
  - Bagging for data.
  - Sampling for features.
- In the random forest algorithm, we do NOT need to do pruning for each "weak" decision tree.
- Advantage of Random Forest
  - Less overfitting
  - parallel implementation
- Random Forest vs. Bagging
  Difference at the feature sampling step.

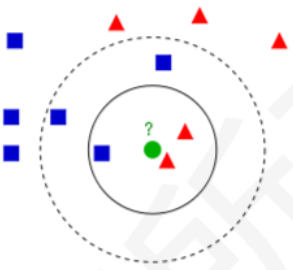
3.4. Feature Importance value in Random Forest

$$importance(i) = performance(RF) - performance(RF^{(random\ i)})$$

It does not signify positive or negative impact on the class label. It only judges how much class discriminatory information each variable contains.

4. K-nearest-neighbors

4.1. Model introduction



- K denotes the k labels that are "closest" to your target sample.
  The higher K is, the smoother (more robust) your model can be.

- Multiple definition choices for "distance"
  - CityBlock Distance, Euclidean Distance.

4.2. Implement KNN algorithm

Find the top K shortest distance from your returned unsorted distance array. (随后python coding - heap sort内容会详细讲解.)

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4.3. FLANN annoy faiss

locality-sensitive hashing
https://github.com/spotify/annoy